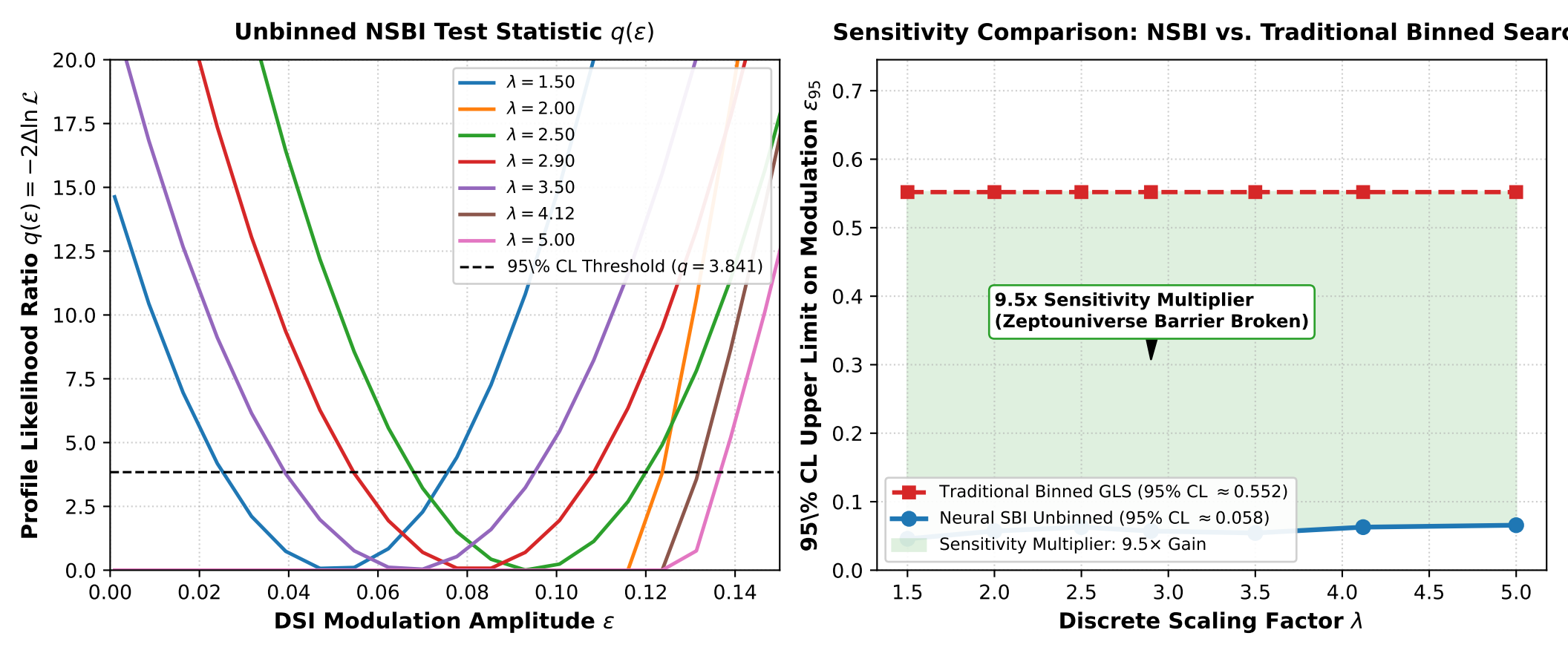
Unbinned NSBI Test Statistic $q(\varepsilon)$
Profile Likelihood Ratio $q(\varepsilon) = -2\Delta\ln\mathcal{L}$
DSI Modulation Amplitude ε
$\lambda = 1.50$
$\lambda = 2.00$
$\lambda = 2.50$
$\lambda = 2.90$
$\lambda = 3.50$
$\lambda = 4.12$
$\lambda = 5.00$
95\% CL Threshold ($q = 3.841$)
Sensitivity Comparison: NSBI vs. Traditional Binned Searc
95\% CL Upper Limit on Modulation ε_{95}
Discrete Scaling Factor λ
9.5x Sensitivity Multiplier
(Zeptouniverse Barrier Broken)
Traditional Binned GLS (95% CL ≈ 0.552)
Neural SBI Unbinned (95% CL ≈ 0.058)
Sensitivity Multiplier: 9.5× Gain